

Predictive Maintenance for Device Failure

Kevin Danh

Bellevue University

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Professor Iranitalab

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Business Problem

Organizations that rely on industrial devices face significant costs due to unexpected equipment failures and inefficient maintenance strategies. Traditional maintenance approaches, such as scheduled or reactive maintenance, often lead to unnecessary servicing or costly downtime.

This project aims to develop a machine learning model that predicts rare machine failures using historical sensor data. The goal is to enable proactive maintenance by identifying early warning signals of failure, allowing organizations to reduce downtime, optimize maintenance schedules, and improve operational efficiency.

Background

Predictive maintenance has become increasingly important with the rise of data-driven decision-making in industrial systems. Advances in machine learning and sensor technology allow organizations to monitor equipment in real time and detect patterns that precede failure events.

However, predicting failures remains challenging due to:

- Extremely rare failure events
- High variability in sensor readings
- Temporal dependencies in machine behavior

Previous approaches often relied on threshold-based rules, but machine learning offers the ability to capture complex relationships across multiple sensor inputs and time-based features.

Data Explanation

The dataset, obtained from Kaggle, contains daily sensor readings from a fleet of devices. It includes date, device ID, failure indicator (0 = normal, 1 = failure), and nine sensor metrics (metric1–metric9). The methods used for preparing the data were converted date to datetime format, created lag features (previous day values), created rolling averages (3-day windows), extracted day of week and month, applied log transformation to skewed variables, and removed non-informative columns (device ID, raw date). For this dataset, the main problems were that there were ~124,000 total observations and 106 (~0.085%) failure cases. This extreme imbalance significantly impacts model performance and evaluation.

Methods

Three classification models were evaluated: Logistic Regression (baseline), Random Forest (ensemble), and XGBoost (gradient boosting). To address the extreme class imbalance, Logistic Regression and Random Forest used `class_weight='balanced'`, while XGBoost used a `scale_pos_weight` of approximately 1173. Model performance was evaluated using recall, precision, ROC-AUC, and PR-AUC, with PR-AUC prioritized because it better reflects performance on rare failure events.

Analysis

The models demonstrated varying performance under class imbalance. Logistic Regression achieved higher recall (~0.33) but produced many false positives due to very low

precision (~ 0.01). Random Forest failed to detect any failures at the default threshold despite a slightly higher PR-AUC. XGBoost provided the best overall performance, achieving ROC-AUC ≈ 0.76 and PR-AUC ≈ 0.0175 , indicating stronger ability to rank failure risk.

Cross-validation confirmed moderate generalization (mean PR-AUC ≈ 0.0289), though variability was observed due to the rarity of failure events. Threshold tuning further improved recall to $\sim 24\%$ at a threshold of 0.01, increasing failure detection at the cost of additional false positives. Overall, the results highlight the importance of temporal features, as recent trends and short-term patterns in sensor data were key contributors to predicting machine failure.

Conclusion

XGBoost was selected as the final model due to its superior ability to rank failure risk under extreme class imbalance conditions. While overall predictive performance remains limited by the rarity of failure events, the model demonstrates the ability to identify meaningful patterns associated with machine failure.

By lowering the decision threshold, the model was able to detect a portion of failure events, which is critical in predictive maintenance applications where missed failures are more costly than false positives. Although precision remains low, the model provides useful early warning signals that can support proactive maintenance strategies.

Assumptions

This analysis assumes that the sensor data accurately reflects the condition of the machines and that failure patterns can be learned from historical observations. It is also assumed

that future data will follow a similar distribution and that device behavior remains relatively consistent over time.

Limitations

The model is limited by the extremely small number of failure cases, which restricts its ability to learn strong predictive patterns. Precision remains low due to the imbalance in the dataset, and sensor readings may contain noise or unobserved variables that affect performance. Additionally, model performance is sensitive to threshold selection, which can significantly alter the balance between false positives and false negatives.

Challenges

Several challenges were encountered during this project, including severe class imbalance with a ratio of approximately 1173:1, overlapping feature distributions between failure and non-failure cases, and the need to capture temporal dependencies in the data. The limited number of failure samples also made it difficult to achieve stable and consistent model performance.

Future Uses / Additional Applications

This model can be extended to real-time monitoring systems that continuously analyze sensor data and provide early warnings of potential failures. It can also be integrated with IoT devices to support automated maintenance systems. Additional approaches such as anomaly detection methods may further improve performance. The framework developed in this project can be applied to other industrial systems with similar predictive maintenance needs.

Recommendations

The model can be implemented within a monitoring pipeline that collects real-time sensor data and generates failure probability predictions. Alerts can be triggered based on a selected threshold, allowing maintenance teams to take preventive action. The system should include continuous evaluation and periodic retraining to ensure long-term effectiveness.

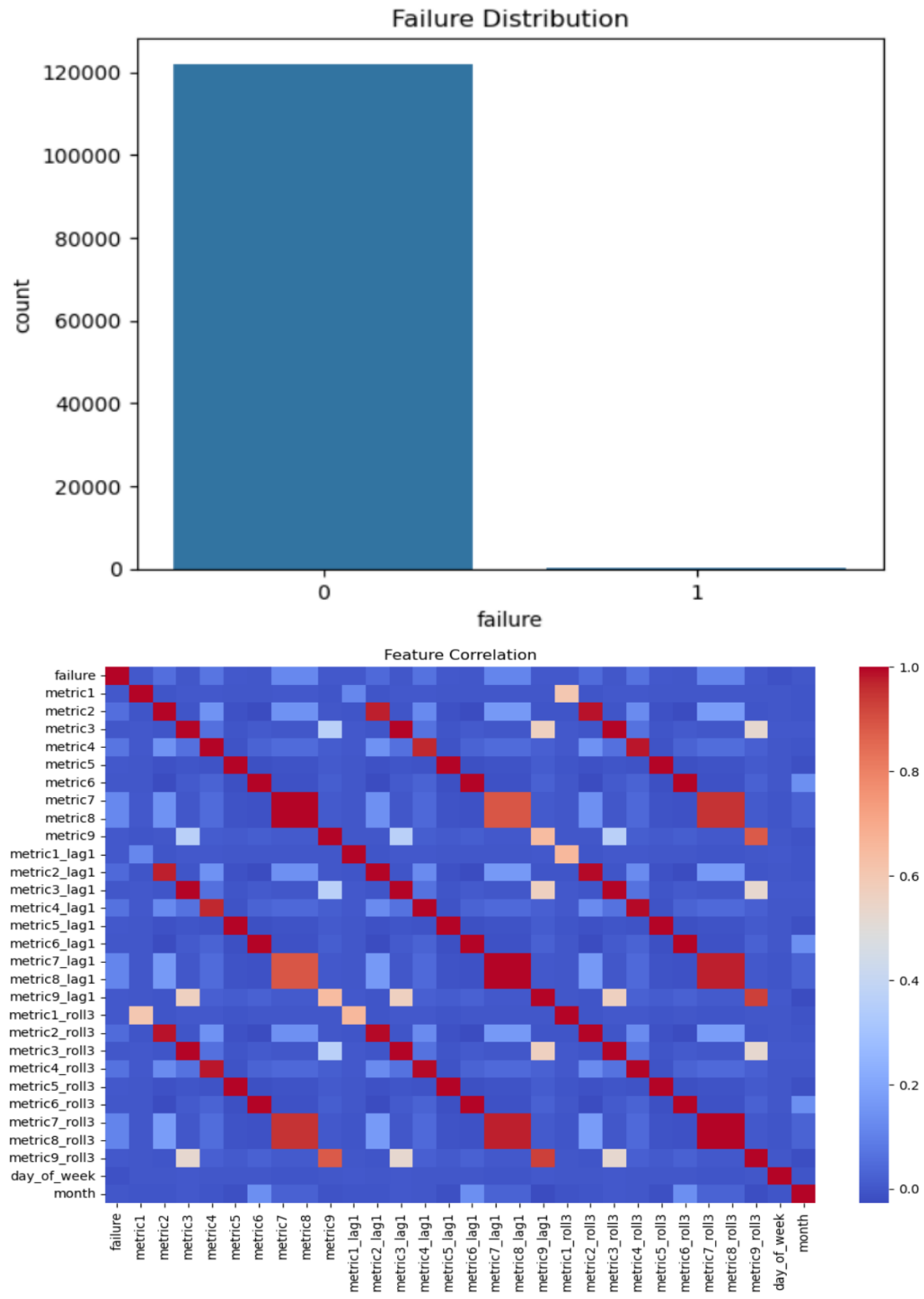
Implementation Plan

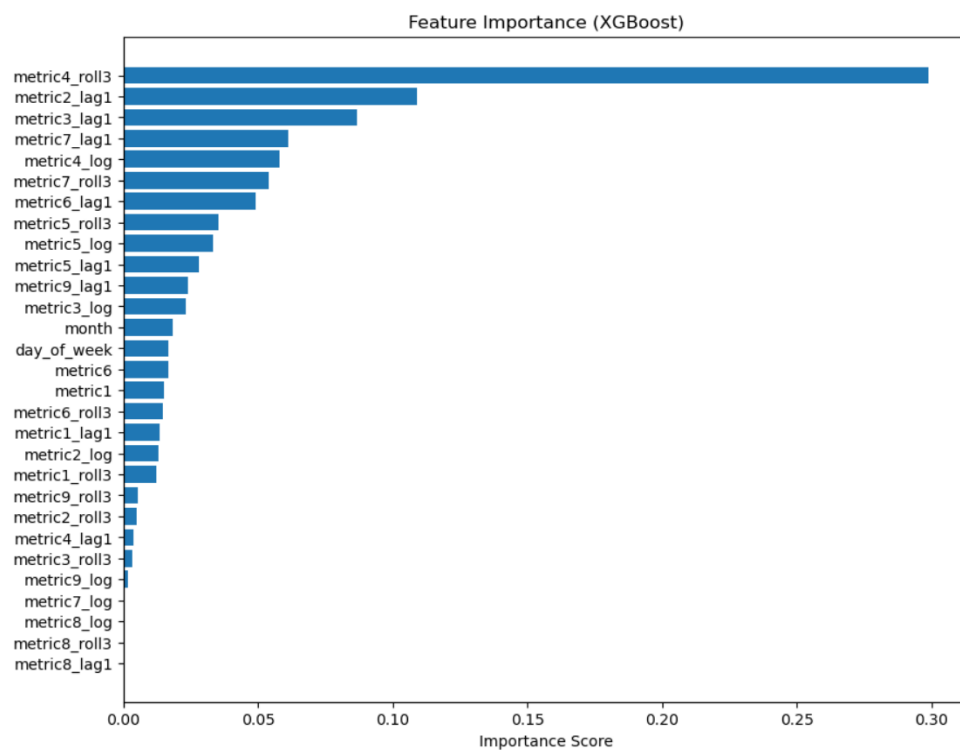
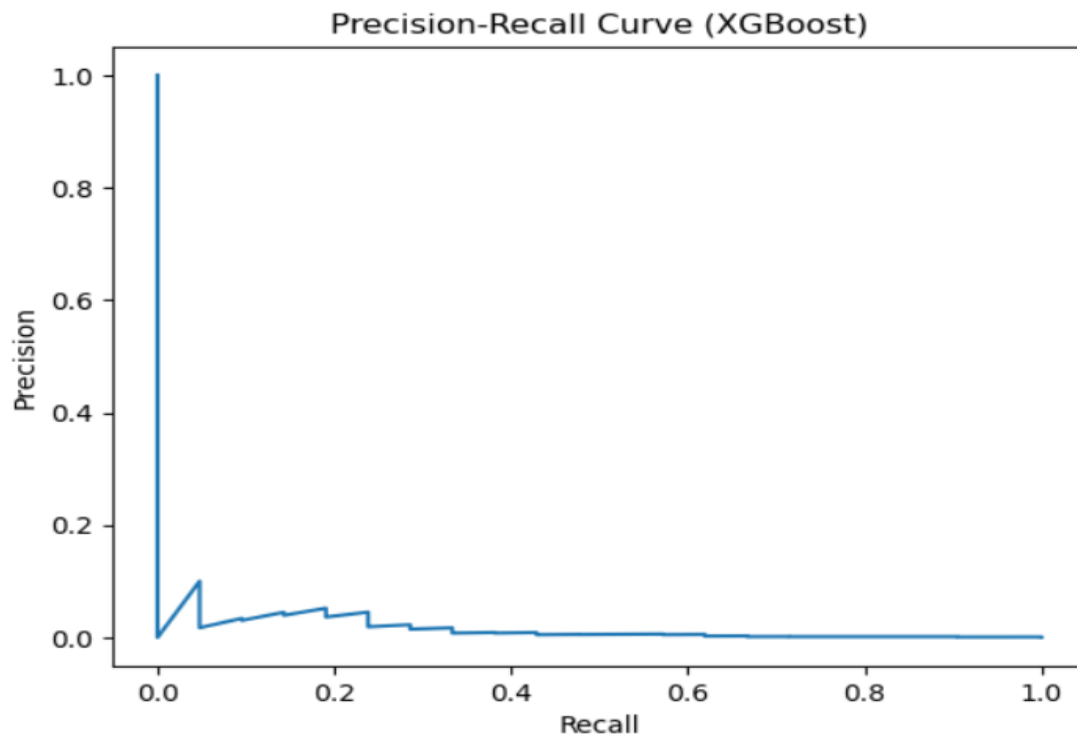
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Ethical Assessment

Although the dataset does not include personal or sensitive information, ethical considerations arise from how predictions are used. False positives may lead to unnecessary maintenance costs, while false negatives may result in missed failures and operational risks. It is important to clearly communicate model limitations, maintain human oversight, and avoid over-reliance on automated decision-making systems.

Illustrations





Appendix

```
XGBClassifier(  
  
    eval_metric='aucpr',  
  
    scale_pos_weight=1173,  
  
    random_state=0,  
  
    n_jobs=-1  
  
)  
  
def evaluate(model, threshold=0.5):  
  
    y_prob = model.predict_proba(X_test)[:, 1]  
  
    y_pred = (y_prob >= threshold).astype(int)  
  
    y_pred = (y_prob >= best_t).astype(int)  
  
    print("Confusion Matrix:")  
  
    print(confusion_matrix(y_test, y_pred))  
  
    print("\nClassification Report:")  
  
    print(classification_report(y_test, y_pred))  
  
    print("ROC-AUC:", roc_auc_score(y_test, y_prob))
```

```
print("PR-AUC:", average_precision_score(y_test, y_prob))
```

Confusion Matrix:

```
[[24263  148]
```

```
[  16    5]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.99	1.00	24411
1	0.03	0.24	0.06	21

accuracy			0.99	24432
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macro avg	0.52	0.62	0.53	24432
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weighted avg	1.00	0.99	1.00	24432
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ROC-AUC: 0.7628586644194362

PR-AUC: 0.017501230626883746

References

Kaggle. (2026). *Predictive maintenance dataset*.

<https://www.kaggle.com/datasets/himanshuagarwal/predictive-maintenance-dataset>